Stellar Cyber 6.6.0s is now live! Discover all the updates in the release notes!

Virtual Modular Sensor Provisioning and Performance

This article describes the resources required for the successful installation and operation of Stellar Cyber virtual modular sensors. Virtual modular sensors install as dedicated VMs with Stellar Cyber software pre-installed.

Refer to the following sections for details:

- **What Determines Sensor Performance?**
- **Sensor Provisioning Rules**
- **Sensor Provisioning by Module**

What Determines Sensor Performance?

Virtual modular sensor performance depends on the following key factors:

- Allocated VM resources (CPU and Memory). Sensor performance scales with CPU and RAM.
- Enabled features (Log Forwarder, Network Traffic, Sandbox, and IDS).
- Traffic volume (both peak and sustained events per second, connections per second, and bandwidth usage).

More features, deeper inspection, and more traffic always require more CPU and RAM.

Sensor Provisioning Rules

Guarantee your virtual modular sensor performance by following these rules for sensor provisioning:

- Reserve CPU and memory resources in the hypervisor.
- Enable SS3 for Linux-based sensors.
- Use ≥2.1 GHz processors for validated performance.

- Provision at least 1.5 times as much memory as the number of CPU cores to ensure stable performance. So for example, if you provision **8** CPUs, you should provision at least **8 x 1.5 = 12GB** of memory.
- Provision sensor disk space using the rules in **Provisioning Disk Space for Virtual Modular Sensors**.
- Stellar Cyber limits the number of log sources you can point at a given sensor as follows:
    - Sensors with 8 GB or less of memory accept a maximum of 100 log sources.
    - Sensors with more than 8 GB of memory accept a maximum of 200 log sources.


Log sources beyond the limit are aggregated under the special source IP address of 0.0.0.0.


As a simple rule, more traffic and inspection depth require more CPU and memory.

Sensor Provisioning by Module

Stellar Cyber provides virtual modular sensors using specific modules. This section lists the provisioning you must provide to achieve the performance for your purchased module. See the sections below for details.


Stated specifications support the corresponding maximum network traffic inspection throughput. Performance may vary depending on your environment, configuration, and other variables.

Network-Module Sensor Specifications

The Network-Module sensor specifications assume a span port, local file assembly, packet forwarding, log forwarding, and Network Traffic. The Malware Sandbox and IDS features are **not enabled**.

Virtual machines for the Network-Module sensors must be provisioned as follows:

Component	Network-Module-1G	Network-Module-10G
Virtual Cores (reserved)	8	32
RAM (GB) (reserved)	16	64
Maximum throughput (Gbps) (1 or 2 span ports)	Up to 1.0	Up to 10.0
Log Forwarder Events/s	7,000	10,000

Security-Module Sensor Specifications – IDS Only

The Security-Module IDS sensor specifications assume a span port, local file assembly, log forwarding, Network Traffic, and IDS. The Malware Sandbox feature is **not enabled**.

Virtual machines for the Security-Module sensors must be provisioned as follows:

Component	Security-Module-200M IDS	Security-Module 500M IDS	Security-Module 1G IDS	Security-Module 10G IDS
Virtual Cores (reserved)	4	8	16	32
RAM (GB) (reserved)	8	16	64	64
Maximum throughput (Mbps) (1 or 2 span ports)	Up to 200	Up to 500	Up to 1,000	Up to 10,000
Log Forwarder Events/s	1,500	1,500	7,000	10,000

Security-Module Sensor Specifications – IDS and Sandbox

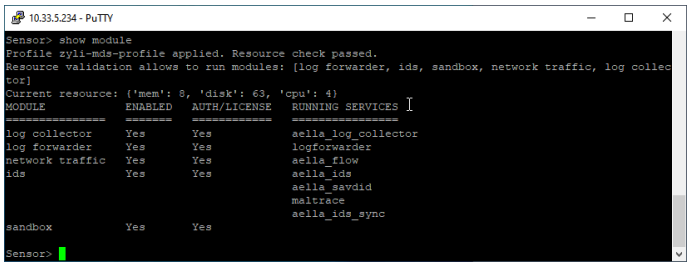
The Security-Module sensor specifications assume a span port, local file assembly, log forwarding, Network Traffic, Malware Sandbox, and IDS.

Virtual machines for the Security-Module IDS sensors must be provisioned as follows:

Component	Security-Module-200M	Security-Module-500M	Security-Module-1G	Security-Module-10G
Virtual Cores (reserved)	4	8	16	32
RAM (GB) (reserved)	8	16	64	64
Maximum throughput (Mbps) (1 or 2 span ports)	Up to 200	Up to 500	Up to 1,000	Up to 10,000
Log Forwarder Events/s	1,500	1,500	7,000	10,000

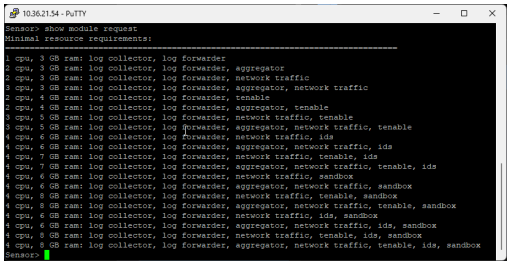
Checking Enabled Features on the Sensor

You can see which features are currently enabled on a sensor, as well as its current provisioning by using the **show module** command in the Sensor CLI. For example:

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Stellar Cyber Provisioning Recommendations

You can see the most recent provisioning guidance from Stellar Cyber by using the **show module request** command in the Sensor CLI. For example:



Automatic Resource Checking

When you apply a Modular Sensor Profile, Stellar Cyber automatically checks the CPUs and RAM provisioned for the target sensor and informs you whether they are sufficient to support the features enabled in the profile. You can also see the results in the **show module** command listed above – the output of the command reports both the name of the Modular Sensor Profile applied and whether the resource checks passed.

If the resource checks do not pass, Stellar Cyber informs you of the issue and displays the sensor in the list with a visible warning of the problem. It does not enable the sensor features until the minimum specifications are met.



Stellar Cyber only checks the CPUs and RAM against the **show module request** recommendations. Disk size checks are not enforced.

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